G9-BLOWUP-LIMIT-RIGIDITY-AUDIT-v1: Blow-up limit + rigidity, NOT W_+ estimate

ZFL-CX

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1 Frozen state

$\boxed{G7 = FROZEN,\ G8 = CLOSED/BLOCKED,\ E\text{-}K = NO\ REOPEN}$ $\boxed{\mathcal{C}_{\text{PROVEN}} = \{C_E^{weak}, C_{L^3}, C_F^{conc}, C_I^{freq}}$

Bottleneck: $\boxed{\int_0^{T^*} W_+ = +\infty}$ not controlled by audited classes.

2 New class: NEXT CLASS = BLOW-UP LIMIT + RIGIDITY

No more W_+ estimate. No more critical norm estimate. New question: $\boxed{\text{What must EVERY blow-up limit satisfy?}}$

3 Entry rule - 6 steps, kill immediate if unprovable

1. **Sequence**: Define $r_n \to 0$, $(x_n, t_n) \to (x^*, T^*)$, $u^{(n)}(y, s) = r_n u(x_n + r_n y, t_n + r_n^2 s)$, $s \in [-1, 0]$. What choice of (x_n, t_n, r_n) is forced by $\mathcal{C}_{\text{PROVEN}}$? No Type I assumption $r \sim (T^* - t)^{1/2}$. r_n must come from $C_F^{conc} + C_I^{freq}$ only.

2. **Compactness**: What convergence can we prove for $u^{(n)}$? Aubin-Lions from C_E^{weak} gives L^2_{loc}, but need L^3_{loc} to keep C_{L^3}? Local energy inequality survives? p? Identify exact compactness without assuming bound on $\|\nabla u^{(n)}\|_2$ uniform beyond scaling.

3. **Limit equation**: Limit $\bar{u}$ solves ancient NS or Euler? $\nu_n = \nu r_n^{-?}$? If $r_n \to 0$, viscous term $\nu_n = \nu$ after scaling? Actually $u^{(n)}$ solves NS with same ν if r_n scaling is Leray. Euler if $\nu r_n \to 0$ not same scaling. Determine precisely.

4. **Surviving information**: Which of $C_{L^3}, C_F^{conc}, C_I^{freq}, C_J^{neg}$ survives to $\bar{u}$? C_{L^3} gives $\|\bar{u}\|_{L^3} \lesssim 1$? Or measure? C_F^{conc} gives non-zero? C_J^{neg} gives something about direction? Must be derived, not assumed.

5. **Rigidity**: Is there independent Liouville theorem killing $\bar{u}$ given surviving info? E.g., ancient mild NS solution with $\|u\|_{L^3} < \infty$ and concentration $\Rightarrow 0$? Known: Escauriaza-Seregin-Sverak L^3 ancient $\Rightarrow 0$ under $u \in L_t^\infty L_x^3$. Does our $\bar{u}$ satisfy hypotheses? Need exact statement. No fabrication.

6. **Nontriviality**: Need $\bar{u} \neq 0$ forced by blow-up assumption. C_F^{conc} should give that, but must be quantitative: $\int_{B_1} |\bar{u}|^3 \geq c_0 > 0$. If not provable, KILL.

If any step requires hypothesis not in $\mathcal{C}_{\text{PROVEN}}$ or not provable from $T^* < \infty \Rightarrow \boxed{KILL\ immediate}$.

4 What is NOT allowed

- Assume $\bar{\alpha} > 0$ (killed by shear barrier $\bar{u} = (f(y), 0, 0)$, $\bar{\alpha} \equiv 0$). - Assume $r_n \sim \sqrt{T^* - t_n}$ (Type I, blocked in G8-A). - Assume $\int |\bar{\omega}|^2 \bar{\alpha} \geq c_0$ (that's G8-B killed). - Use $D_t e_i$ alignment (G8-C blocked). - Use $\|u\|_3 \|\nabla\omega\|_2^2$ as bridge. - Use CKN as direct bridge.

5 Kill criteria for this audit

KILLED if: - compactness requires bound not in $\mathcal{C}_{\text{PROVEN}}$, - limit equation requires $\nu_n \to 0$ unjustified, - surviving info insufficient for any known Liouville, - nontriviality $c_0 > 0$ not provable, - rigidity theorem needs smallness/regularity not derived.

6 Decision

This audit only answers: does there exist a definable sequence, compactness, limit equation, surviving info, and independent rigidity that together are all provable from $T^* < \infty$?

If NO: $\boxed{G9\text{-}DESIGN = CLOSED/BLOCKED}$, NEXT CLASS also exhausted, no.tex of actual proof. If YES: then sole authorized G9-01 can be designed with explicit mechanism + kill criteria.

$\boxed{\text{This is NOT G9-01, only mechanism check}}$